Primary 3 · Science

Science · Set 3

80 marks · 1 h 15 min · Exam mock

Name: Class: Score: / 80

Section A: Multiple-Choice Questions (40 marks)

Choose the correct answer. Each question is worth 2 marks.

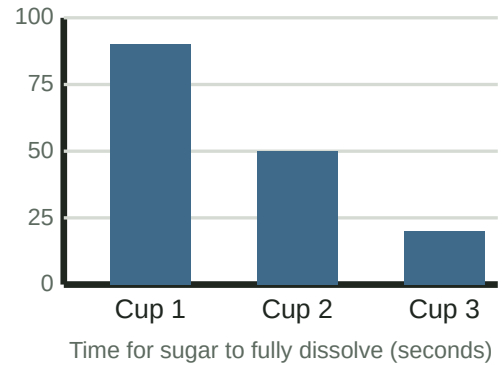
1. A pupil tests how fast sugar dissolves in three cups of water. Cup 1 has cold water, Cup 2 has warm water, and Cup 3 has hot water. She puts the same amount of sugar in each cup and stirs each one the same number of times. What was CHANGED between the three cups? ( )

(1) Water temperature (2) Amount of sugar (3) Number of times stirred
(4) Type of cup

2. In the same test, what was KEPT THE SAME for every cup? ( )

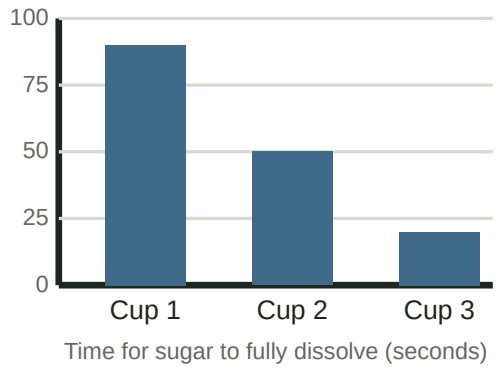
(1) The amount of sugar and number of stirs (2) The temperature of the water
(3) How long she waited for it to dissolve (4) The colour of the water

3. The graph shows how long the sugar took to dissolve in Cup 1 (cold), Cup 2 (warm) and Cup 3 (hot) water. Based on the graph, which cup dissolved the sugar FASTEST? ( )



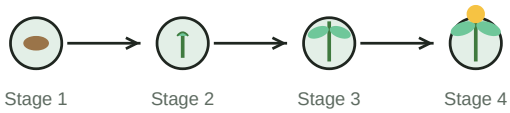
(1) Cup 1 (2) Cup 2 (3) All took the same time (4) Cup 3

4. What does the bottom of this graph show was changed between the three cups? ( )



(1) The temperature of the water (2) The amount of sugar used
(3) The number of times stirred (4) The size of the cup

5. Look at the stages of a bean plant's life cycle. Which stage comes right BEFORE the "seedling" stage? ( )



(1) Seed (2) Young plant (3) Adult plant with flowers (4) Fruit

6. For a seed to germinate (start to grow), which of these does it need? ( )

(1) Sunlight and soil only (2) Water and air (3) A flower pot (4) Fertiliser

7. Which of these is a living thing? ( )

(1) Statue (2) Mushroom (3) Raincoat (4) Bicycle

8. Which group of animals is usually covered in fur and feeds its young with milk? ( )

(1) Birds (2) Fish (3) Insects (4) Mammals

9. Which of these animals hatches from an egg? ( )

(1) Guinea pig (2) Rabbit (3) Lizard (4) Cat

10. Which of these is TRUE about bacteria? ( )

(1) They can be seen easily without a microscope (2) All bacteria cause disease
(3) Bacteria can make their own food using sunlight
(4) Some bacteria are helpful, e.g. in making yoghurt

11. Which of these animals' life cycle has NO pupa stage? ( )

(1) Butterfly (2) Beetle (3) Mosquito (4) Cockroach

12. A ladybird's life cycle has 4 stages: egg, larva, pupa, adult. Which other animal in this list also has 4 stages? ( )

(1) Cockroach (2) Grasshopper (3) Mosquito (4) Chicken

13. Which stage of a butterfly's life cycle is when it does NOT eat and its body changes completely? ( )

(1) Egg (2) Caterpillar (larva) (3) Pupa (4) Adult

14. Which body part does a tadpole lose as it grows into an adult frog? ( )

(1) Tail (2) Legs (3) Eyes (4) Mouth

15. Which material is flexible (can bend without breaking)? ( )

(1) Glass (2) Ceramic (3) Rubber (4) Dry twig

16. Which of these objects would float in water? ( )

(1) A steel spoon (2) An empty plastic bottle (3) A stone (4) An iron nail

17. Which material is see-through (transparent)? ( )

(1) Wood (2) Cardboard (3) Cloth (4) Clear plastic

18. Which of these is a non-living thing? ( )

(1) Ant (2) Tree (3) Rainbow (4) Bird

19. Which animal is covered in scales and lives mostly in water? ( )

(1) Frog (2) Fish (3) Duck (4) Rabbit

20. Mushrooms get their food by... ( )

(1) Making it using sunlight like plants (2) Breaking down dead or decaying matter
(3) Absorbing it from the air directly as a gas (4) Photosynthesis in their roots

Section B: Structured Questions (16 marks)

Write a short answer, or tick True or False, for each question. Each question is worth 2 marks.

21. Tick True or False for each statement. Write True or False.

Sentence	True / False
(a) Giving a plant fertiliser is a way to TEST whether it is alive.	
(b) Watching a plant's leaves slowly turn to face the sunlight is a way to TEST whether it is alive.	

22. A pupil tests how far three identical toy cars roll after being pushed with the same force, once on a tile floor, once on a carpet, and once on grass. What was CHANGED in this test? ( )

(1) The surface the car rolls on (2) The toy car used (3) The force used to push it
(4) The person doing the test

23. Give one property that makes plastic suitable for making the handle of a cooking pot. (2 marks)

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24. Give one property that makes rubber suitable for making a rubber band. (2 marks)

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25. Which of these is a way fungi are DIFFERENT from green plants? ( )

(1) Fungi can reproduce (2) Fungi need water
(3) Fungi cannot make their own food using sunlight (4) Fungi can grow

26. Tick True or False for each statement. Write True or False.

Sentence	True / False
(a) A beetle's life cycle has 4 stages, including a pupa.	
(b) A grasshopper's life cycle has a pupa stage.	

27. Tick True or False for each statement. Write True or False.

Sentence	True / False
(a) A material that is strong does not tear or break easily.	
(b) A material that is waterproof allows water to pass through it easily.	

28. Besides needing food, name ONE other characteristic shared by ALL living things. (2 marks)

Section C: Open-Ended Questions (24 marks)

Explain your answer in full sentences. Each question is worth 3 marks.

29. A pupil wants to find out which cloth — cotton, paper towel or felt — absorbs the most water. She cuts each cloth to the same size, dips each one into the same amount of water for the same length of time, then weighs how much water each cloth has absorbed. Describe what was changed, what was kept the same, and why this makes it a fair test. (3 marks)

30. A pupil places identical ice cubes into three containers made of different materials — metal, plastic and styrofoam — and leaves them at the same room temperature. She records how long each ice cube takes to melt completely. Describe what was changed, what was kept the same, and why this makes it a fair test. (3 marks)

31. Describe TWO tests you could carry out to find out whether a small animal you found in the garden is alive, without just looking at it. (3 marks)

32. In MacRitchie forest, mushrooms often grow on a fallen, rotting log. Explain how the mushrooms get their food from the log, and what eventually happens to the log because of this. (3 marks)

33. A cockroach's life cycle has only 3 stages (egg, nymph, adult), while a beetle's life cycle has 4 stages (egg, larva, pupa, adult). Explain ONE difference between these two life cycles. (3 marks)

34. A pupil wants to choose a material to make a raincoat. Explain why plastic would be a good choice, giving TWO properties of plastic that make it suitable. (3 marks)

35. Hafiz says that mushrooms and the moss growing nearby are the same kind of living thing because both grow in damp, shady places. Explain the KEY difference between fungi (like mushrooms) and plants (like moss). (3 marks)

36. A pupil plants two identical seeds. She keeps Seed A moist but keeps Seed B completely dry. After a week, Seed A has sprouted but Seed B has not changed at all. Explain why this happened. (3 marks)

End of paper. Check your answers.